

PHASE 4

PROJECT PLANNING

POCKETSMART AI — Smart Budget & Recommendation Assistant

Expanded phase evidence with detailed wording, diagrams, workflow, and project-focused explanation.

Planning the build

Project planning organizes the work into manageable stages so the idea can become a functioning application. The project is planned around a reusable common flow rather than building three unrelated systems. First the team establishes authentication and navigation, then planner inputs, recommendation generation, result presentation, History, testing, and final demonstration.

Roadmap thinking

The roadmap starts with understanding the problem and continues through requirements, design, development, testing, documentation, and demonstration. Each phase produces evidence for the next. For example, requirement analysis identifies the inputs needed by Home, Party, and Jewelry planners; design turns those requirements into routes, models, templates, and service boundaries; development implements them; testing checks the expected behaviour.

Budget-planning perspective

Budget planning is treated as a primary decision variable throughout the application. The recommendation result is expected to show a budget summary and estimated allocation rather than only listing products. This helps users understand how the available amount is being distributed across categories and what balance remains.

Visual Model



Figure — Phase visual model

Process Flow



Figure — Detailed workflow / roadmap

PHASE 4 — DETAILED CONTENT

PocketSmart AI connects practical planning with a structured recommendation workflow. The following section expands the phase in clear project language so the document has sufficient explanatory depth for academic/project review.

Project Explanation

A strong budget-planning assistant should reduce uncertainty without hiding the reasoning context behind the result. In PocketSmart AI, the user supplies the information that matters for the selected category, including budget and relevant preferences. The application then processes that context and presents a structured recommendation. This structure makes the output easier to read, compare, and explain during a project demonstration. The project also demonstrates how a common architecture can support different use cases. Home Interior planning focuses on rooms, furniture, lighting, fans, and dining requirements. Party planning focuses on guests, event type, venue type, catering, decoration, and entertainment. Jewelry planning focuses on occasion, style preferences, budget, and optional outfit-image context. Although the inputs differ, the overall application pattern remains consistent.

Key Concepts

Workstream	Planned focus
Product scope	Home, Party, Jewelry, Dashboard, History.
Backend	FastAPI routes, models, authentication, Gemini utility.
Frontend	Jinja2 templates and HTML/CSS result pages.
Security	Hashing, HTTP-only cookies, environment-based API key.
Delivery	Testing evidence, documentation, and demonstration flow.

Roadmap / Mind-Map View

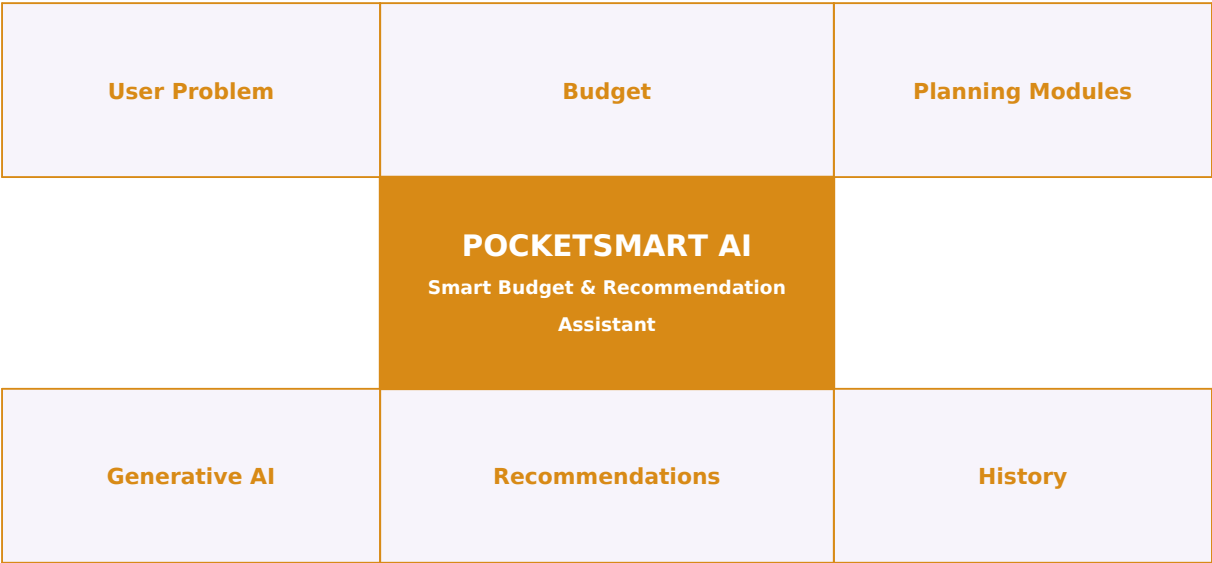


Figure — Additional project roadmap visual

Project Evidence

The supplied project documentation includes interface and workflow evidence. A representative screenshot is included here to connect the written explanation with the implemented application.

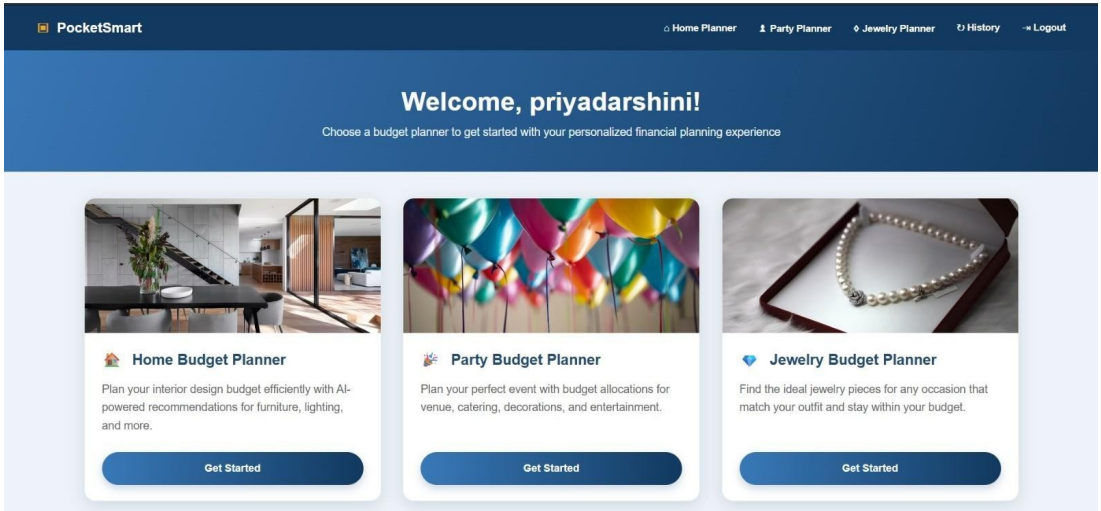
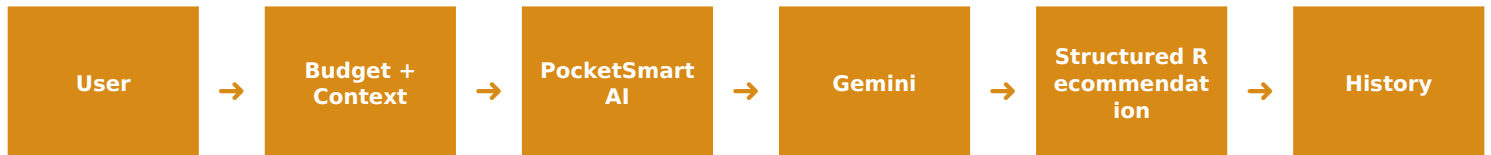


Figure — Representative PocketSmart AI project evidence

PHASE SUMMARY

Phase 4 contributes to the overall PocketSmart AI lifecycle by making the project easier to understand, build, validate, or present. The emphasis in this phase is on clear communication: the reader should be able to see what the system is doing, why the step matters, and how it connects to the final budget-aware recommendation workflow.

End-to-End Connection



Project Reflection

PocketSmart AI is designed around a practical user problem: making planning decisions while respecting a defined budget. The project brings together web development, authentication, structured data, Generative AI, visual presentation, and History. This combination makes the application a useful example of how a software project can move from a simple idea to a demonstrable end-to-end system. The documentation should therefore communicate not only the features, but also the reasoning, workflow, and evidence behind them.

Closing Note

The phase is complete when its purpose, workflow, supporting evidence, and connection to PocketSmart AI are understandable to a reviewer without requiring them to infer missing steps. The expanded paragraphs and diagrams in this version are intended to make that explanation clearer and more professional.